

Market context (Sep 27)

- **WTI:** \$92.41. **Brent futures:** about \$104.
- **Drivers:** US–Iran truce talks versus Houthi strikes on Saudi Arabia.
- **How much WTI has actually moved (annualized):** 46% over the last 20 days, 43% over 60 days, 63% since the war began in March. Source: CL_Futures sheet in Aarin's repo.
- **AAA gas:** \$4.48 today, flat week over week (\$4.476 a week ago). This matches the lag model, which said the pump would stop rising after crude's \$9 drop.

Research finding: Kalshi's WTI ladder prices the far wings inconsistently

I backed out the volatility each Dec 31 strike implies, using a standard lognormal (GBM) model with WTI at \$92.41 and 94 days left. For "touch" contracts, $P(\text{touch } B) \approx 2 \times P(\text{finish above } B)$.

KXWTIMAX strike (touch) \$115 \$125 \$140 \$160 \$180 \$200

Mid price	34¢	19¢	13.8¢	7.8¢	6.5¢	5.8¢
Implied vol	45%	45%	55%	61%	71%	80%

KXWTIMIN strike (touch) \$65 \$64 \$62 \$60 \$55

Mid price	15.5¢	17¢	10.5¢	6.5¢	4¢
Implied vol	49%	53%	49%	46%	50%

- **The \$125 strike and the \$60 strike are the cheapest volatility on the ladder**, priced at about 45–46%, which is the calmest recent realized level.
- **The \$200 strike is priced as if volatility were 80%**, higher than WTI has actually moved even during the war.
- **KXWTIMIN T64 costs more than T65.** Nothing in the ladder explains that, and the same "retail flow creates mispricings" point shows up in the other team's deck.

The trades (per \$2,069 of capital, 1,000 contracts each)

Leg	Contract	Side	Limit	Cost	Max gain	Role
A	KXWTIMAX-26DEC31-T125	YES	$\leq 20¢$	\$199	\$801	Upside tail (escalation)
B	KXWTIMIN-26DEC31-T60	YES	$\leq 8¢$	\$80	\$920	Downside tail (truce)
C	KXWTIMAX-26DEC31-T200	NO	$\leq 95¢$	\$950	\$50	Sell the overpriced far wing
D	KXAAAGASMAX-26DEC31-5.40	NO	$\leq 84¢$	\$840	\$160	Carry from the stationary gas margin

Open interest is 294k, 27k, 332k and 61k contracts respectively. Use limit orders. B's spread is wide (5¢ bid / 8¢ ask).

Leg A: YES, WTI touches \$125 by Dec 31

- **Model probability:** 18.6% at 45% vol, 27.9% at 55%, 34.5% at 63%.
- **Price:** 19.9¢. This is fair if WTI calms down and cheap if the war regime continues.
- **Why:** it's the escalation half of the strangle. Houthi strikes on Saudi infrastructure are the trigger, and Brent already hit \$126 in March.

Leg B: YES, WTI touches \$60 by Dec 31

- **Model probability:** 5.9% at 45% vol, 12.2% at 55%, 17.7% at 63%.
- **Price:** 8¢.
- **Why the model understates it:** a US–Iran deal is a jump, not a random walk. Before the war WTI traded \$57–67 (Dec 2025: \$57.26; Feb 2026: \$66.96), so a truce that removes the war premium makes \$60 realistic.
- **Why T60:** it's the cheapest downside strike on the ladder (46% implied vol).

Leg C: NO, WTI touches \$200

- **Model probability of YES:** 0.1–1.6% at any volatility from 45% to 63%. The market charges 5–6.5¢.
- **EV:** $(1 - 0.016) - 0.95 \approx +3$ to **+5¢ per contract**, about 4% in 3 months.
- **Only loses if:** WTI doubles, meaning a full Strait of Hormuz closure. In that scenario Leg A pays \$801.

Leg D: NO, AAA gas reaches \$5.40 by Dec 31

- **What it needs:** +92¢ from \$4.48.
 - At today's \$2.18 retail margin, that requires WTI sustained at about \$135: $(5.40 - 2.18) \times 42$. The market prices a touch of \$135 at only about 16%, and gas needs weeks above that level, not a touch.
 - All-time record is \$5.02 (Jun 2022).
- **Why it should revert:**
 - The retail margin (pump minus crude cost) is stationary, with a 4.6-week half-life. It's now 2.2σ above its 5-year average of \$1.62.
 - Seasonal: gas fell from Sep 20 to year-end in every year from 2022 to 2025, by 18–78¢ (average –49¢).
- **Estimate:** YES fair $\approx$ 8–10% versus 16–19¢ on the market, so $EV \approx +6$ to **+8¢ per contract**.
- **Why \$5.40 and not \$5.00:** at \$5.00 (30¢), gas could reach the strike with WTI at only about \$118, before Leg A pays. At \$5.40, gas can't get there unless WTI has already passed \$125.

Scenario P&L (1,000 of each leg)

Scenario	A	B	C	D	Total
Truce: WTI touches \$60	–199	+920	+50	+160	+931
Chop: WTI stays \$65–120	–199	–80	+50	+160	–69

Scenario	A	B	C	D	Total
Escalation: WTI touches \$125–135, gas stays under \$5.40	+801	–80	+50	+160	+931
Bigger escalation: WTI above \$135, gas hits \$5.40	+801	–80	+50	–840	–69
Extreme: WTI touches \$200	+801	–80	–950	–840	–1,069

The shape is a long straddle. A plain CME straddle loses its whole premium if nothing happens. Here, the two NO legs earn \$210, which covers 75% of the \$279 spent on the tails.

Expected value

Leg	At 45% vol	At 55% vol
A	–\$13	+\$80
B	–\$21	+\$42
C	+\$49	+\$44
D	+\$70 (model-independent)	+\$70
Total	+\$85	+\$236

Kalshi fees $\approx 0.07 \times P \times (1 - P)$ per contract, about \$28 total for the book.

Risks

1. **Extreme tail:** WTI at \$200 or gas at \$5.40 without a WTI spike are the only paths to a –\$1,069 loss. Both need a Hormuz-scale event or a refinery shock.
2. **Volatility collapse:** if a slow, grinding truce keeps WTI between \$70 and \$110, you lose \$69.
3. **Settlement definitions:** KXWTIMAX and KXWTIMIN settle on the front-month WTI futures settlement price, and AAA gas on the AAA daily national average. The model inputs use futures and EIA data. Confirm both rulebooks before trading.
4. **Model risk:** the lognormal model has no jumps. The truce and escalation cases are exactly the jumps it misses, which cuts in the book's favor on A and B and against it on C.
5. **Liquidity:** B's spread is 3¢ wide. Scale in with limit orders.

Management rules

- **Take profits:** sell half of A if YES trades above 50¢ (roughly WTI above \$115). Sell half of B if YES trades above 30¢ (roughly WTI below \$72).
- **Cut D:** if AAA gas goes above \$4.95, or WTI closes above \$120 for 3 days in a row.
- **Add the natural gas leg:** once Kalshi lists the October KXNATGASMON contracts, buy NO on strikes 25–30% away from spot (spot is about \$3.20 now), but only if YES trades at 12¢ or more. Empirically those moves happen 6–9% of the time in a month, because gas mean-reverts with a 4.4-week half-life. Stop before mid-November, when winter spike risk starts; Henry Hub spot hit \$30.72 in Jan 2026.

- **Review weekly:** re-run Aarin's scraper on KXWTIMAX, KXWTIMIN and KXAAAGASMAX and recompute the implied volatilities.